Status	Finished
Started	Wednesday, 28 October 2026, 6:48 PM
Completed	Wednesday, 28 October 2026, 6:50 PM
Duration	2 mins 3 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are discarded
- c. They are compared element by element
- d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting helps in finding duplicates quickly
- c. Sorting makes it harder to search for items
- d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **3**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It complicates the analysis process
- c. It has no effect on the analysis process
- d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **4**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Selection
- b. Searching
- c. All of the above
- d. Duplicates

The correct answer is: All of the above

Question **5**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. arrange()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **6**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Arranging data in a particular format
- c. Deleting data from a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **7**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Linear Search
c. Quick Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Merge Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort is less efficient for large lists
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Makes data harder to manage
c. Decreases the efficiency of algorithms
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. sorted()
c. order()
d. sort()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By breaking the input into smaller parts and merging them
c. By using the bubble-up method
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **13**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Complexity

c. Searching

d. Sorting

The correct answer is:
Complexity

Question **14**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 80,150,200,470,40,90,300,400,70,120

b. 40,70,80,90,120,150,200,300,400,470

c. 200,470,80,150,40,90,300,400,70,120

d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time

b. It compares adjacent elements and swaps them if necessary

c. It divides the list into two halves, sorts each half, and then merges them

d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Status	Finished
Started	Wednesday, 28 October 2026, 6:32 PM
Completed	Wednesday, 28 October 2026, 6:41 PM
Duration	8 mins 42 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums`, sort the array in **increasing** order based on the frequency of the values. If multiple values have the same frequency, sort them in **decreasing** order.

Print the *sorted array*.

Example 1:

```
Input:
6
1 1 2 2 2 3

Output:
3 1 1 2 2 2

Explanation: '3' has a frequency of 1, '1' has a frequency of 2, and '2' has a frequency of 3.
```

Example 2:

```
Input:
5
2 3 1 3 2
Output:
1 3 3 2 2
Explanation: '2' and '3' both have a frequency of 2, so they are sorted in decreasing order.
```

Example 3:

```
Input:
9
-1 1 -6 4 5 -6 1 4 1
Output:
5 -1 4 4 -6 -6 1 1 1
```

Constraints:

- `1 <= nums.length <= 100`
- `-100 <= nums[i] <= 100`

For example:

Input	Result
6 1 1 2 2 2 3	3 1 1 2 2 2
5 2 3 1 3 2	1 3 3 2 2

Answer: (penalty regime: 0 %)

```python
import sys
from collections import Counter

def main():
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    n = int(input_data[0])
    nums = [int(x) for x in input_data[1:n+1]]

    freq = Counter(nums)

    sorted_nums = sorted(nums, key=lambda x: (freq[x], -x))

    print(*(sorted_nums))

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	6 1 1 2 2 2 3	3 1 1 2 2 2	3 1 1 2 2 2	✓
✓	5 2 3 1 3 2	1 3 3 2 2	1 3 3 2 2	✓
✓	9 -1 1 -6 4 5 -6 1 4 1	5 -1 4 4 -6 -6 1 1 1	5 -1 4 4 -6 -6 1 1 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Objective:

Develop a Python program to find the k-th maximum value in a given list of integers. The program should handle various edge cases, including lists with duplicate values, empty lists, and invalid values of k. The k-th maximum value refers to the k-th largest distinct element in the list.

Background:

Finding the k-th maximum value in a list is a common problem in computer science, often encountered in fields like data analysis, competitive programming, and software development. This problem requires an understanding of sorting algorithms, data structures, and efficient problem-solving techniques. By solving this problem, one gains insights into how to handle large datasets and optimize performance in practical applications.

Problem Description:

Given a list of integers, the task is to determine the k-th maximum value in the list. The program should meet the following requirements:

1. **Input:**
    - A list of integers, which can contain both positive and negative values.
    - An integer k, representing the position of the maximum value to find.
2. **Output:**
    - The k-th maximum value in the list.
    - If k is greater than the number of distinct elements in the list or if the list is empty, the program should return an appropriate message indicating the error.

Constraints:

- The list may contain duplicate values.
- The value of k should be a positive integer.
- The list may contain up to 10^6 elements, and each element can be as large as 10^9 in magnitude.

Examples:

Consider the following examples for better understanding:

1. **Example 1:**
    - **Input:** list = [3, 1, 5, 4, 2], k = 2
    - **Output:** 4
    - **Explanation:** The distinct elements in the list are [1, 2, 3, 4, 5]. The 2nd maximum value is 4.
2. **Example 2:**
    - **Input:** list = [7, 7, 7, 7, 7], k = 1
    - **Output:** 7
    - **Explanation:** The distinct elements in the list are [7]. The 1st maximum value is 7.
3. **Example 3:**
    - **Input:** list = [2, 1, 2, 1, 2], k = 3
    - **Output:** -1
    - **Explanation:** The distinct elements in the list are [1, 2]. There is no 3rd maximum value.

For example:

Input	Result
5 3 1 5 4 2 2	4
6 7 7 7 7 7 7 1	7
10 2 1 2 1 2 1 2 1 2 1 3	-1

Answer: (penalty regime: 0 %)

```python
import sys

def main():
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    n = int(input_data[0])
    nums = [int(x) for x in input_data[1:n+1]]
    k = int(input_data[n+1])

    distinct_sorted = sorted(list(set(nums)), reverse=True)

    if k <= 0 or k > len(distinct_sorted):
        print("-1")
    else:
        print(distinct_sorted[k - 1])

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	5 3 1 5 4 2 2	4	4	✓
✓	6 7 7 7 7 7 7 1	7	7	✓
✓	10 2 1 2 1 2 1 2 1 2 1 3	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given an integer array `nums` sorted in **non-decreasing** order, return *an array of **the squares of each number** sorted in non-decreasing order*.

Example 1:

```
Input: nums = [-4,-1,0,3,10]
Output: [0,1,9,16,100]
Explanation: After squaring, the array becomes [16,1,0,9,100].
After sorting, it becomes [0,1,9,16,100].
```

Example 2:

```
Input: nums = [-7,-3,2,3,11]
Output: [4,9,9,49,121]
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 <= nums[i] <= 10^4`
- `nums` is sorted in **non-decreasing** order.

For example:

Test	Result
`print(sortedSquares([-4,-1,0,3,10]))`	`[0, 1, 9, 16, 100]`

Answer: (penalty regime: 0 %)

Reset answer

```
def sortedSquares(nums):
    return sorted(x * x for x in nums)
```

	Test	Expected	Got	
✓	`print(sortedSquares([-4,-1,0,3,10]))`	`[0, 1, 9, 16, 100]`	`[0, 1, 9, 16, 100]`	✓
✓	`print(sortedSquares([-7,-3,2,3,11]))`	`[4, 9, 9, 49, 121]`	`[4, 9, 9, 49, 121]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

To find the frequency of numbers in a list and display in sorted order.

Constraints:

1<=n, arr[i]<=100

Input:

1 68 79 4 90 68 1 4 5

output:

1 2

4 2

5 1

68 2

79 1

90 1

For example:

Input	Result
4 3 5 3 4 5	3 2 4 2 5 2

Answer: (penalty regime: 0 %)

```
arr = list(map(int, input().split()))
freq = {}

for num in arr:
    freq[num] = freq.get(num, 0) + 1

for num in sorted(freq):
    print(num, freq[num])
```

	Input	Expected	Got	
✓	4 3 5 3 4 5	3 2 4 2 5 2	3 2 4 2 5 2	✓
✓	12 4 4 4 2 3 5	2 1 3 1 4 3 5 1 12 1	2 1 3 1 4 3 5 1 12 1	✓

	Input	Expected	Got	
✓	5 4 5 4 6 5 7 3	3 1 4 2 5 3 6 1 7 1	3 1 4 2 5 3 6 1 7 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Given an listof integers, sort the array in ascending order using the *Bubble Sort* algorithm above. Once sorted, print the following three lines:

1. List is sorted in numSwaps swaps., where numSwaps is the number of swaps that took place.
2. First Element: firstElement, the *first* element in the sorted list.
3. Last Element: lastElement, the *last* element in the sorted list.

For example, given a worst-case but small array to sort: a=[6,4,1]. It took 3 swaps to sort the array. Output would be

```
Array is sorted in 3 swaps.
```

```
First Element: 1
```

```
Last Element: 6
```

Input Format

The first line contains an integer,n , the size of the list a .
The second line contains n, space-separated integers a[i].

Constraints

· 2<=n<=600

· 1<=a[i]<=$2x10^6$.

Output Format

You must print the following three lines of output:

1. List is sorted in numSwaps swaps., where numSwaps is the number of swaps that took place.
2. First Element: firstElement, the *first* element in the sorted list.
3. Last Element: lastElement, the *last* element in the sorted list.

Sample Input 0

3

1 2 3

Sample Output 0

List is sorted in 0 swaps.

First Element: 1

Last Element: 3

For example:

Input	Result
3 3 2 1	List is sorted in 3 swaps. First Element: 1 Last Element: 3
5 1 9 2 8 4	List is sorted in 4 swaps. First Element: 1 Last Element: 9

Answer: (penalty regime: 0 %)

```
n = int(input())
a = list(map(int, input().split()))

numSwaps = 0

for i in range(n):
    swaps = 0
    for j in range(n - 1):
        if a[j] > a[j + 1]:
            a[j], a[j + 1] = a[j + 1], a[j]
```

```
            numSwaps += 1
            swaps += 1
    if swaps == 0:
        break

print("List is sorted in", numSwaps, "swaps.")
print("First Element:", a[0])
print("Last Element:", a[-1])
```

	Input	Expected	Got	
✓	3 3 2 1	List is sorted in 3 swaps. First Element: 1 Last Element: 3	List is sorted in 3 swaps. First Element: 1 Last Element: 3	✓
✓	5 1 9 2 8 4	List is sorted in 4 swaps. First Element: 1 Last Element: 9	List is sorted in 4 swaps. First Element: 1 Last Element: 9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

The problem is that we want to reverse a array in O(N) linear time complexity and we want the algorithm to be in-place as well!
For example: input is [1,2,3,4,5] then the output is [5,4,3,2,1]
Input
5
1 2 3 4 5
Output
5 4 3 2 1

For example:

Input	Result
5 1 2 3 4 5	5 4 3 2 1

Answer: (penalty regime: 0 %)

```
n = int(input())
arr = list(map(int, input().split()))

arr.reverse()

print(*arr)
```

	Input	Expected	Got	
✓	5 1 2 3 4 5	5 4 3 2 1	5 4 3 2 1	✓
✓	10 0 2 4 6 8 1 3 5 7 9	9 7 5 3 1 8 6 4 2 0	9 7 5 3 1 8 6 4 2 0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array of integers `arr`, replace each element with its rank.

The rank represents how large the element is. The rank has the following rules:

- Rank is an integer starting from 1.
- The larger the element, the larger the rank. If two elements are equal, their rank must be the same.
- Rank should be as small as possible.

Example 1:

```
Input: arr = [40,10,20,30]
Output: [4,1,2,3]
Explanation: 40 is the largest element. 10 is the smallest. 20 is the second smallest. 30 is the third smallest.
```

Example 2:

```
Input: arr = [100,100,100]
Output: [1,1,1]
Explanation: Same elements share the same rank.
```

Example 3:

```
Input: arr = [37,12,28,9,100,56,80,5,12]
Output: [5,3,4,2,8,6,7,1,3]
```

Constraints:

- $0 <= arr.length <= 10^5$
- $-10^9 <= arr[i] <= 10^9$

For example:

Test	Result
print(arrayRankTransform([40,10,20,30]))	[4, 1, 2, 3]

Answer: (penalty regime: 0 %)

Reset answer

```
def arrayRankTransform(arr):
    sorted_arr = sorted(set(arr))
    rank = {}

    for i, num in enumerate(sorted_arr):
        rank[num] = i + 1

    return [rank[num] for num in arr]
```

	Test	Expected	Got	
✓	`print(arrayRankTransform([40,10,20,30]))`	`[4, 1, 2, 3]`	`[4, 1, 2, 3]`	✓
✓	`print(arrayRankTransform([100,100,100]))`	`[1, 1, 1]`	`[1, 1, 1]`	✓
✓	`print(arrayRankTransform([37,12,28,9,100,56,80,5,12]))`	`[5, 3, 4, 2, 8, 6, 7, 1, 3]`	`[5, 3, 4, 2, 8, 6, 7, 1, 3]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given an integer array `nums`, return *an integer array* `counts` *where* `counts[i]` *is the number of smaller elements to the right of* `nums[i]`.

Example 1:

```
Input: nums = [5,2,6,1]
Output: [2,1,1,0]
Explanation:
To the right of 5 there are 2 smaller elements (2 and 1).
To the right of 2 there is only 1 smaller element (1).
To the right of 6 there is 1 smaller element (1).
To the right of 1 there is 0 smaller element.
```

Example 2:

```
Input: nums = [-1]
Output: [0]
```

Example 3:

```
Input: nums = [-1,-1]
Output: [0,0]
```

Constraints:

- `1 <= nums.length <= 10^5`
- `-10^4 <= nums[i] <= 10^4`

For example:

Test	Result
`print(countSmaller([5,2,6,1]))`	`[2, 1, 1, 0]`
`print(countSmaller([-1]))`	`[0]`

Answer: (penalty regime: 0 %)

Reset answer

```
def countSmaller(nums):
    sorted_nums = sorted(set(nums))
    rank = {num: i + 1 for i, num in enumerate(sorted_nums)}
    bit = [0] * (len(rank) + 1)
    counts = []

    def query(i):
        total = 0
        while i > 0:
            total += bit[i]
            i -= i & -i
        return total

    def update(i):
        while i < len(bit):
            bit[i] += 1
            i += i & -i

    for num in reversed(nums):
        r = rank[num]
        counts.append(query(r - 1))
        update(r)

    return counts[::-1]
```

	Test	Expected	Got	
✓	`print(countSmaller([5,2,6,1]))`	`[2, 1, 1, 0]`	`[2, 1, 1, 0]`	✓
✓	`print(countSmaller([50,20,60,10]))`	`[2, 1, 1, 0]`	`[2, 1, 1, 0]`	✓
✓	`print(countSmaller([-1]))`	`[0]`	`[0]`	✓
✓	`print(countSmaller([-1, -1]))`	`[0, 0]`	`[0, 0]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Bubble Sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements if they are in wrong order. You read an list of numbers. You need to arrange the elements in ascending order and print the result. The sorting should be done using bubble sort.

Input Format: The first line reads the number of elements in the array. The second line reads the array elements one by one.

Output Format: The output should be a sorted list.

For example:

Input	Result
6 3 4 8 7 1 2	1 2 3 4 7 8
5 4 5 2 3 1	1 2 3 4 5

Answer: (penalty regime: 0 %)

```
n = int(input())
arr = list(map(int, input().split()))

for i in range(n):
    for j in range(n - 1 - i):
        if arr[j] > arr[j + 1]:
            arr[j], arr[j + 1] = arr[j + 1], arr[j]

print(*arr)
```

	Input	Expected	Got	
✓	6 3 4 8 7 1 2	1 2 3 4 7 8	1 2 3 4 7 8	✓
✓	6 9 18 1 3 4 6	1 3 4 6 9 18	1 3 4 6 9 18	✓
✓	5 4 5 2 3 1	1 2 3 4 5	1 2 3 4 5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Write a Python program to sort a list of elements using the merge sort algorithm.

For example:

Input	Result
5 6 5 4 3 8	3 4 5 6 8

Answer: (penalty regime: 0 %)

```
def merge_sort(arr):
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])

    return result


n = int(input())
arr = list(map(int, input().split()))

arr = merge_sort(arr)
print(*arr)
```

	Input	Expected	Got	
✓	5 6 5 4 3 8	3 4 5 6 8	3 4 5 6 8	✓
✓	9 14 46 43 27 57 41 45 21 70	14 21 27 41 43 45 46 57 70	14 21 27 41 43 45 46 57 70	✓
✓	4 86 43 23 49	23 43 49 86	23 43 49 86	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.